

Thermal material for faster computing

GE recently came up with a phase-change based prototype substrate can be applied to computer chips for higher heat dissipation

Google's clean power tower

Google's invests \$168 million in a solar power tower which will generate a large amount of clean solar energy

Tomorrow's Tech

We're talking Terahertz here

Factors limiting how 'fast' computers can get also include other problem occurring out of scaling down transistors to such levels. Have you noticed that processor frequency hasn't gone up for quite some time? Sure, you have hyper-threading and multiple cores, but why the ceiling on clock speed?

One of the reasons is that the transistors themselves need to get faster, but can't anymore. When we say that transistors need to become faster, we mean the rate at which they switch. As a thumb rule, the thinner the transistor gate, the higher is the electric field created (as is evinced in the working of a capacitor). The higher the strength of the electric field the faster the transistor would switch. But right now the gate isolator underneath the gate on a 45 nm process measures 0.9nm in thickness, which is just a little larger than a single silicon-dioxide molecule. Most believe it can't get smaller than this.

As Jonathan Kang, CPU Architect,

peculiar electronic properties Graphene intrinsically exhibits. Graphene is essentially a sheet of carbon just an atom thick (or thin). So thin that you can think of it as two dimensional! Graphene also allows for much higher rates of electron flow. Last year, IBM showcased Graphene transistors that were capable of switching at clock frequencies of 100 GHz. These transistors were produced using techniques similar to those used in producing silicon wafers, so commercial production is possible. Digital signals are still a problem for these transistors though, so for now, use in conventional computing is ruled out. "Because graphene is a zero bandgap semiconductor, meaning there is no energy difference between its conductive and nonconductive states, transistors made of graphene cannot be turned on and off. In contrast, silicon has a bandgap of one electron volt, making it good for processing discrete digital signals", Yu-Ming Lin, the lead researcher is reported to have stated. As work continues though, scientists are hopeful of encountering newer breakthroughs.

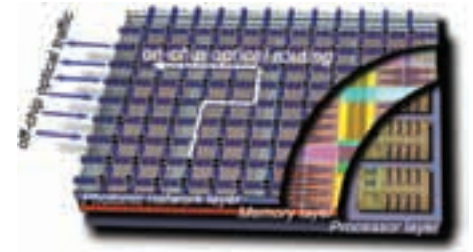
Recently, the same research group announced taking the cut of frequency of graphene transistors to 155 GHz. When work in this area was first published in 2008, researchers hoped that THz Graphene transistors could be achieved in an optimised graphene transistor with a gate length of 50 nanometers. The gate length of the latest ones were 40 nm. So graphene has been a little slow

to come through, but will likely make transistors bad-ass eventually. But what about the other slow pokes?

The inevitable bottleneck

When one microcomponent operates faster than the surrounding ecosystem, it will obviously not help overall computing become faster. Bottlenecks can

happen within the component itself or the surrounding components in the process chain. They can even happen because of the bus transport bandwidth. Take the processor; if it can spew out information at a certain number of cycles per second and your memory doesn't have the bandwidth to accept information at such a rate, there's a bottleneck. The



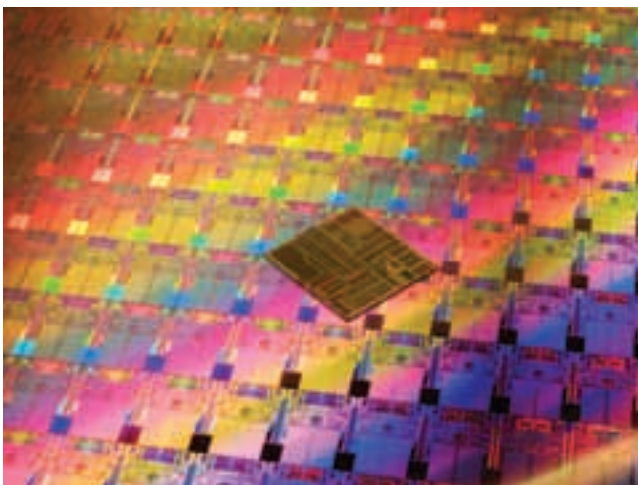
Top layer featuring photonics network

front side bus also comes into play here.

To understand bottlenecks, think of computing as a large congested metropolis. Building fly-overs in such a city only helps you get to your traffic jam faster. It won't help you get to your destination quicker. So far we've been talking about transistors and the roadblocks limiting their scaling. However, another issue has to do with the chip or IC containing these little transistors. The interconnects between these transistors are the limiting factor now. Architecturally band-aid solutions have been found for this problem but as computers traverse Teraflops, to Petaflops to even Exaflops of compute power, things will start to clogging. What is needed, is a faster and more efficient on and off chip connectivity option.

Let there be light

One of the most promising technologies that aims to remove on-chip bottlenecks in preparation for the Exaflop era is IBM's CMOS Silicon Nanophotonics technology. If IBM has its way, in the near future, chips will be communicating using light pulses instead of electrical charge. The premise is simple – if optics can replace copper over longer distances of communication maybe they can even become competitive over shorter and shorter distances. The ultimate goal of this project is to develop



The Penryn Wafer photographed with an original Intel Pentium processor die. The new processors incorporate 820 million transistors for each quad core chip

puts it on Quora.com, while explaining these difficulties, "The simple matter is that the easy scaling we've had in the past where every shrink in transistor size would also lead to faster transistors is over". Scientists are hence looking towards newer materials.

Graphene seems to be the most likely choice here. This is because of the